

JACKSON, MN, USA

WMO: 726593

Lat: 43.650N

Lon: 94.986W

Elev: 1446

StdP: 13.94

Time Zone: -6.00 (NAC)

Period: 98-19

WBAN: 04946

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.1	-5.9	-17.0	2.3	-8.4	-12.8	2.9	-5.5	34.5	14.1	31.9	14.8	9.6	310	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	90.1	73.3	87.7	71.8	83.7	69.8	77.6	85.1	75.2	82.4	73.2	80.4	13.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.3	139.9	81.0	72.8	128.2	79.7	71.6	123.1	78.8	42.2	84.8	39.8	82.4	37.8	79.8	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
28.4	25.3	22.6		-16.4	95.0	5.5	3.4	-20.3	97.4	-23.5	99.4	-26.6	101.3	-30.6	103.8	
				-17.1	80.7	5.8	3.4	-21.3	83.2	-24.7	85.2	-27.9	87.1	-32.1	89.6	
				DB												
				WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.2	17.1	20.5	32.9	46.3	58.1	68.2	71.9	68.8	62.6	48.8	35.6	21.5	
	DBStd	21.74	12.82	12.42	12.37	10.15	9.19	6.73	5.92	5.76	8.55	9.75	10.72	11.96	
	HDD50	4086	1020	826	546	186	30	0	0	0	9	141	445	883	
	HDD65	7508	1485	1246	997	565	252	43	11	26	144	510	882	1347	
	CDD50	2683	0	0	15	73	282	547	680	583	386	104	13	0	
	CDD65	632	0	0	0	3	39	140	226	144	72	8	0	0	
	CDH74	5732	0	0	5	57	449	1263	2036	1163	665	92	2	0	
	CDH80	1693	0	0	0	12	137	405	645	298	175	21	0	0	
Wind	WSAvg	10.4	12.0	12.0	11.8	12.3	11.6	9.7	7.5	6.9	8.5	10.3	11.5	11.4	
Precipitation	PrecAvg	30.3	0.7	0.7	1.7	3.2	4.1	4.8	3.9	3.7	3.0	2.2	1.3	0.9	
	PrecMax	45.3	1.5	1.8	3.1	6.5	6.8	10.2	7.7	9.8	8.7	6.5	4.1	2.4	
	PrecMin	20.3	0.1	0.0	0.1	1.0	1.3	2.3	0.7	0.7	0.4	0.3	0.1	0.1	
	PrecStd	6.1	0.4	0.4	0.9	1.4	1.6	1.9	1.8	2.0	2.3	1.6	1.0	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.1	54.5	72.3	81.3	89.9	91.4	93.0	91.0	90.1	82.0	69.8	54.0	
		MCWB	39.1	46.7	56.2	59.8	66.3	71.0	76.2	77.3	69.4	65.5	53.9	49.4	
	2%	DB	39.2	46.0	63.1	73.2	83.6	88.4	90.1	86.4	85.6	75.1	61.4	45.8	
		MCWB	35.2	41.9	52.7	55.1	63.3	69.0	75.7	74.5	69.5	61.1	51.2	40.2	
	5%	DB	36.4	39.4	55.1	69.5	79.2	84.4	87.6	82.3	81.2	70.1	55.2	39.4	
		MCWB	34.2	36.6	48.9	54.0	61.4	67.3	74.5	71.7	66.3	57.4	46.3	35.3	
	10%	DB	33.9	36.3	50.3	63.2	73.3	81.2	82.4	80.8	78.5	64.0	51.5	36.3	
		MCWB	32.5	34.4	45.4	50.9	58.6	66.3	71.2	70.3	64.8	54.3	44.9	34.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.9	49.5	61.6	62.4	71.3	75.5	81.5	81.8	75.2	72.5	57.1	50.1
			MCDB	45.6	55.1	68.8	74.2	82.3	85.1	88.1	87.7	83.9	77.9	65.2	53.7
2%		WB	36.5	42.1	54.7	58.5	67.4	73.2	78.6	77.1	72.4	63.3	52.3	40.3	
		MCDB	39.1	46.4	59.5	69.0	77.6	82.9	85.8	83.0	80.8	68.4	59.9	45.1	
5%		WB	34.2	37.3	50.1	55.7	64.4	71.0	76.1	74.4	70.0	59.2	48.0	36.6	
		MCDB	36.1	39.7	56.2	65.9	73.9	79.8	83.9	80.2	77.2	66.9	54.8	39.4	
10%		WB	32.2	34.3	44.7	53.0	61.7	68.9	74.1	72.1	67.4	55.4	44.3	34.0	
		MCDB	33.4	36.0	49.8	60.8	69.9	77.5	81.3	78.4	74.8	63.6	50.5	36.0	
Mean Daily Temperature Range	5% DB	MDBR	15.1	15.5	15.8	19.8	20.6	19.7	19.5	19.3	21.9	19.8	16.8	14.3	
		MCDBR	17.1	18.5	25.1	29.4	27.4	24.4	21.1	21.2	25.2	27.4	24.1	19.0	
		MCWBR	14.0	14.3	15.9	15.4	12.3	10.3	11.0	11.7	12.0	14.4	14.7	14.4	
	5% WB	MCDBR	15.5	17.6	23.4	24.2	22.3	19.3	19.0	18.6	21.4	21.9	22.7	17.6	
		MCWBR	13.1	13.8	15.6	14.8	12.1	10.8	11.5	11.8	12.4	13.9	14.8	13.8	
Clear-Sky Solar Irradiance	taub	0.267	0.281	0.310	0.359	0.379	0.398	0.410	0.405	0.362	0.323	0.284	0.265		
	taud	2.440	2.419	2.443	2.339	2.350	2.353	2.302	2.295	2.389	2.489	2.522	2.471		
	Ebn at Noon	279	293	296	287	283	277	272	269	274	273	270	269		
	Edh at Noon	23	28	31	38	38	39	40	39	33	26	21	20		
All-Sky Solar Radiation	RadAvg	589	929	1215	1495	1744	1936	2075	1719	1377	918	615	455		
	RadStd	62	99	115	139	124	141	83	125	93	118	48	45		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page